

CENG381 Stochastic Processes

Renewal Processes — Practice 1 (Answer Key)

Q1 Solution

a) Elementary Renewal Theorem and replacement rate:

Elementary Renewal Theorem: for i.i.d. inter-renewal times X with finite mean $E[X]$,

$$N(t)/t \rightarrow 1/E[X] \quad \text{almost surely as } t \rightarrow \infty$$

$$\begin{aligned} \text{Long-run replacement rate} &= 1/E[X] = 1/400 \quad \text{bulbs per hour} \\ &= 0.0025 \quad \text{bulbs/hour} \end{aligned}$$

b) Expected replacements in 8760 hours and $E[X^2]$:

$$E[N(8760)] = 8760 / E[X] = 8760 / 400 = 21.9 \approx 22 \quad \text{bulbs}$$

$$E[X^2] = \text{Var}(X) + (E[X])^2 = 160000 + 160000 = 320000 \quad \text{hours}^2$$

c) Sandwich argument:

Since $S_{\{N(t)\}} \leq t < S_{\{N(t)+1\}}$, dividing through by $N(t)$:

$$S_{\{N(t)\}} / N(t) \leq t / N(t) < S_{\{N(t)+1\}} / N(t)$$

By the SLLN: $S_n / n \rightarrow E[X]$ as $n \rightarrow \infty$

Left side: $S_{\{N(t)\}} / N(t) \rightarrow E[X]$ (as $t \rightarrow \infty$, $N(t) \rightarrow \infty$)

Right side: $S_{\{N(t)+1\}} / N(t) = S_{\{N(t)+1\}} / (N(t)+1) * (N(t)+1) / N(t)$

$$\rightarrow E[X] * 1 = E[X]$$

By the squeeze theorem: $t/N(t) \rightarrow E[X]$

Equivalently: $N(t)/t \rightarrow 1/E[X]$ ✓

Q2 Solution

a) Expected periods and cycle length:

$$E[\text{busy}] = 1/\mu = 1/2 \quad \text{hour}$$

$$E[\text{idle}] = 1/\lambda = 1/3 \quad \text{hour}$$

$$E[\text{cycle}] = E[X] = 1/2 + 1/3 = 3/6 + 2/6 = 5/6 \quad \text{hour}$$

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b) Long-run fraction busy (renewal-reward):

$$E[\text{reward per cycle}] = E[\text{busy period}] = 1/2$$

$$\begin{aligned}\text{Long-run fraction busy} &= E[R] / E[X] = (1/2) / (5/6) \\ &= (1/2) * (6/5) = 3/5 = 60\%\end{aligned}$$

c) CTMC verification:

2-state CTMC: state 1 = Busy, state 0 = Idle. Rate Busy→Idle = $\mu=2$, rate Idle→Busy = $\lambda=3$.

$$Q = \begin{bmatrix} -3, & 3, \\ 2, & -2 \end{bmatrix}$$

$$\pi * Q = 0:$$

$$-3\pi_0 + 2\pi_1 = 0 \Rightarrow \pi_1 = (3/2)\pi_0$$

$$\pi_0 + \pi_1 = 1 \Rightarrow (5/2)\pi_0 = 1 \Rightarrow \pi_0 = 2/5$$

$$\pi_1 = 3/5 \quad \checkmark \quad (\text{matches renewal-reward result})$$

Q3 Solution

a) Turnover rate and expected patients per month:

$$\text{Turnover rate} = 1/E[X] = 1/2 \text{ discharges per day}$$

$$\text{Expected patients in 30 days} = 30 * (1/2) = 15 \text{ patients}$$

b) Time-average residual occupancy (Erlang):

$$Y = E[X^2] / (2 * E[X]) = 6 / (2 * 2) = 6/4 = 3/2 = 1.5 \text{ days}$$

At a random moment, the patient currently occupying the bed is expected to stay for another 1.5 days.

c) Comparison with Exp(1/2) (mean 2 days):

$$\text{For } X \sim \text{Exp}(1/2): \quad E[X] = 2, \quad E[X^2] = 2/\lambda^2 = 2/(1/4) = 8$$

$$Y_{\text{Exp}} = 8 / (2 * 2) = 2 \text{ days}$$

$$Y_{\text{Erlang}} = 1.5 \text{ days} < Y_{\text{Exp}} = 2 \text{ days}$$

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The Erlang(2,1) gives a shorter residual occupancy because it has less variability than the exponential (Var=2 vs Var=4). The inspection paradox is weaker when stays are more regular: the ratio $E[X^2]/E[X] = E[X] + \text{Var}(X)/E[X]$ grows with the coefficient of variation. Regular arrivals reduce the excess wait.